A-A-53981
10 October 1989

COMMERCIAL ITEM DESCRIPTION

CHISEL, DENTAL, BONE, SURGICAL

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification, MIL-C-37741A which is cancelled.

This commercial item description covers dental bone chisels suitable for use in dental surgical procedures.

Salient characteristics:

Classification. The chisels covered by this specification shall be of the following types and numbers as specified below:

Type I. - Chandler:

No. 1
No. 4
No. 7
No. 8

Type II. - Ochsenbein:

No. 1
No. 2

Type III. - Gardner:

No. 5

Materials.

Operative ends. The operative ends shall be fabricated of class 4, type 440A stainless steel. Stainless steel shall be in accordance with ASTM F899 for composition only.

Handles. Handles shall be solid brass with a chrome over nickel plating. Brass shall conform to copper alloy numbers C26000 or C33000 as specified in ASTM B135 for chemical composition only. As an alternate, the handles shall be fabricated of class 3, type 303 or class 4, type 440A stainless steel. Chisels of one piece construction shall be fabricated entirely of 440A stainless steel. Chemical analysis shall be tested as specified herein.

AMSC N/A FSC 6520

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Hardness. The chisels shall be properly stress relieved. The entire instrument for one piece chisels and the operative ends (blades) of two piece instruments shall have a Rockwell hardness of C52 to C56. Handles and shanks of two piece instruments shall be of proper temper and shall have a hardness suitable for the purpose intended. Hardness shall be determined in accordance with ASTM E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Design and dimensions. The bone chisels shall be suitable for use in the removal of bone tissue, or for tooth sectioning, during dental surgical procedures. The chisel shall consist of an operative end, shank and handle. The operative end is the portion that engages the bone tissue or tooth structure. The shank is the portion which extends from the operative end to the handle. The handle is the portion which is held in the hand. The chisel shall be of one or two piece (one piece operative end and shank and one piece handle) construction. The shank shall be permanently secured to the handle. Type I shall be bone chisels of Dr. Chandler's design, numbers 1, 4, 7 and 8. Type II shall be bone chisels of Dr. Ochsenbein's design, numbers 1 and 2. Type III shall be bone chisels of Dr. Gardner's design, number 5. The design and dimensions shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 24371 for type I chisels, Drawing No. 24414 for type II chisels and Drawing No. 24372 for type III chisels. Chisels furnished shall be of the type and number as specified herein.

Finish. Operative ends and shanks shall have a mirror finish, except for the beveled edge, which shall have a fine honed finish. The beveled edge shall have well defined cutting edges. The operative ends shall be free of defects such as coarse grind marks, cracks, burrs, deep pits, feathers, and other imperfections. The gripping portion of the handle shall be serrated or knurled. Handles shall have a mirror or bright finish, except for the knurled or serrated portion. The serrated or knurled portion of the handles shall be free of barbs and provide adequate hand grip during use. The rim edge, used for striking, shall be beveled and tapered to eliminate sharp edges. The interface of the shank and handle shall be smoothly blended with no crevices evident.

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Corrosion resistance. The chisels shall be properly passivated. The copper sulfate test requirements of ASTM F1089 shall apply to the entire instrument of one piece chisels and to the operative end and shank of two piece chisels. The boil test requirements of ASTM F1089 shall apply to the entire chisel.

Identification marking. The chisels shall be marked with the respective chisel number and name of manufacturer. The word "stainless" shall be marked on the stainless steel portion of the instruments. As an alternate to the word "stainless", the use of a registered trade name identifying the steel as stainless shall be acceptable, provided prior approval is received from the cognizant contracting officer. Chisels of foreign manufacture shall be marked with the country of origin.

Workmanship. The operative end shall be free of discolorations and round edges. The chisel shall be finished as specified and shall be free of all defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One chisel, as specified, constitutes one unit.

Unit package. Each chisel shall be wrapped in neutral tissue or other suitable anti-corrosive material and a close fitting container to prevent damage to the operative end.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an inspection level of S-2 for dimensions and II for all other defects. The unit of product for examination purposes shall be one chisel, as specified in this description.

Examination. Examination shall be as specified herein. Examination shall include, but shall not be limited to the following:

Defects

Design not as specified for the applicable type and number chisel.
Cutting edge of operative end not sharp or has rounded edges. *
Operative end and shank not free from coarse grind marks, cracks, burrs, deep pits, or feathers. *
Operative end finish not as specified.
Dimensions not as specified on drawing.
Handle or shank finish not as specified.
Identification marking not present, not properly located, illegible, or incomplete.
Chisel not free from dirt, polishing compound, or foreign matter.
Plating of handle not uniform in appearance.
Plating of handle not free from blisters, visible porosity, or other defects.
Interface of shank and handle not smooth, not complete, or not free of crevices.
Handle serrations missing, not uniform, or not complete.

*Examination shall be under four power (4X) magnification.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified below. The unit of product for test purposes shall be one chisel.

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Test Criteria

Characteristic	Inspection level
Chemical analysis	**
Hardness	S-2
Copper sulfate test	S-2
Boil test	S-2

** A sufficient quantity of each lot of metal used in the construction of the chisels may be selected for composition analysis.

*** Acceptance number shall be zero for all sample sizes.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Chemical analysis. In the absence of certificates of quality, chemical analysis of the individual components of the chisels shall be determined in accordance with method 111 of FED-STD-151.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification	Specification Designation
6520-01-006-9343	CHISEL, DENTAL, Chandler, No. 1, Corrosion-Resisting Steel	Type I
6520-01-006-9344	CHISEL, DENTAL, Chandler, No. 4, Corrosion-Resisting Steel	Type I
6520-01-006-9345	CHISEL, DENTAL, Gardner, No. 5, Corrosion-Resisting Steel	Type III

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6520-01-006-9346	CHISEL, DENTAL, Chandler, No. 7, Corrosion-Resisting Steel	Type I
6520-01-006-9347	CHISEL, DENTAL, Chandler, No. 8, Corrosion-Resisting Steel	Type I
6520-01-047-3585	CHISEL, DENTAL, Ochsenbein, No. 1, Corrosion-Resisting Steel	Type II
6520-01-047-3586	CHISEL, DENTAL, Ochsenbein, No. 2, Corrosion-Resisting Steel	Type II

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2219
LOCATION: NCR DISK 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53981	CHISELS, DENTAL, BONE SURGICAL

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

-

6 REMARKS

7a. NAME OF SUBMITTER *(Last First MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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